

UNIVERSITY OF PISA

DEPARTMENT OF COMPUTER SCIENCE

Master programme in Data Science and Business Informatics

DATA MINING: FUNDAMENTAL

Project report: Seller-Level Vehicle Listings Dataset

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Contents

1 Data Understanding and Preparation **1**
1.1 Data semantics . . . 1
1.2 Distribution of variables and statistics . . . 1
1.2.1 Summary of statistics . . . 1
1.2.2 Distribution via histograms . . . 1
1.2.3 Distribution via pie chart and bar chart . . . 1
1.3 Assessment of data quality . . . 1
1.3.1 Missing value handling . . . 1
1.3.2 Outlier and noise detection . . . 2
1.3.3 Variables transformation . . . 2
1.3.4 Pairwise correlations and variable elimination . . . 2

2 Clustering **2**
2.1 Centroid-based . . . 2
2.2 Density-based . . . 2
2.3 Hierarchical . . . 2
2.4 Final discussion . . . 2

3 Classification **2**
3.1 KNN . . . 2
3.2 Naïve Bayes . . . 2
3.3 Decision Trees . . . 2
3.4 Final conclusion . . . 2

4 Regression **3**
4.1 Univariate regression . . . 3
4.2 Multivariate regression . . . 3

5 Pattern Mining **3**
5.1 Frequent patterns . . . 3
5.2 Association rules . . . 3

6 Conclusion **3**

1 Data Understanding and Preparation

(TODO: introduce the dataset, its source and the overall goal.)

1.1 Data semantics

The Seller-Level Vehicle Listings Dataset contains one record per seller (`sp_id`), summarising the vehicle listings observed for that seller. Describe each variable: meaning, source datatype and measurement scale, as in Table 1. The prediction target is `seller_rating_class` (low / medium / high).

Table 1: Semantic description of the dataset variables. *(Fill in.)*

Variable Name	Description	Datatype	Measurement scale
`sp_id`	Unique seller/dealer identifier	float	Identifier
`n_listings`	Total number of listings	int	Numeric
`price_median`	Median advertised price	float	Numeric
`dominant_body_type`	Most frequent body type	str	Categorical
`franchise_status`	Franchise / independent / mixed / unknown	str	Categorical
`seller_rating_class`	Target rating category	str	Categorical (target)
...	...	...	...

1.2 Distribution of variables and statistics

(TODO: describe how features are distributed and report summary tables and plots.)

1.2.1 Summary of statistics

Table 2 reports count, mean, standard deviation, quartiles and range for the numeric variables.

Table 2: Summary of statistics for numeric variables. *(Fill in from `df.describe()`.)*

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
`price_median`								
`mileage_median`								
`horsepower_median`								
...								

1.2.2 Distribution via histograms

(TODO: insert histograms and discuss skewness and outliers.)

1.2.3 Distribution via pie chart and bar chart

(TODO: insert categorical distributions and the class balance of `seller_rating_class`.)

1.3 Assessment of data quality

1.3.1 Missing value handling

(TODO: report the missing-value heatmap and the imputation strategy per attribute.)

1.3.2 Outlier and noise detection

(TODO: boxplots and the decision on which extreme points to drop or keep.)

1.3.3 Variables transformation

(TODO: encoding of categorical variables and scaling of numeric ones.)

1.3.4 Pairwise correlations and variable elimination

(TODO: correlation heatmap and the reasoning for dropping redundant features.)

2 Clustering

(TODO: describe the clustering setup, the features used, and how dimensionality reduction was applied.)

2.1 Centroid-based

(TODO: K-Means; elbow and silhouette for choosing K.)

2.2 Density-based

(TODO: DBSCAN; k-NN distance plot to choose ε and MinPts.)

2.3 Hierarchical

(TODO: agglomerative clustering; linkage comparison and dendrograms.)

2.4 Final discussion

(TODO: compare the methods and state which worked best.)

3 Classification

The target variable is `seller_rating_class`. Describe the preprocessing, the train/test split, and the evaluation metrics (accuracy, precision, recall, F1, ROC-AUC).

3.1 KNN

(TODO: hyperparameter search results and per-class metrics.)

3.2 Naïve Bayes

(TODO: Gaussian and/or Categorical Naive Bayes results.)

3.3 Decision Trees

(TODO: tuning, pruning, and the resulting tree.)

3.4 Final conclusion

(TODO: compare all classifiers and recommend one.)

Table 3: Model comparison. *(Fill in.)*

Model	Accuracy	Macro-F1	Macro-AUC
k-NN			
Naïve Bayes			
Decision Tree			

4 Regression

(TODO: describe the regression target and predictors.)

4.1 Univariate regression

(TODO: simple regression per predictor.)

4.2 Multivariate regression

(TODO: multiple and multi-target regression.)

5 Pattern Mining

(TODO: describe the discretization/binning of numeric attributes and the transactional representation.)

5.1 Frequent patterns

(TODO: Apriori vs FP-Growth, support and itemset-size analysis.)

5.2 Association rules

(TODO: top rules, support/confidence/lift, and interpretation.)

6 Conclusion

(TODO: summarise the findings and the main takeaways.)

Team Contributions

Member	Contribution
Student One	
Student Two	
Student Three	